

Yes, the original idea exists. Here is what does not.

A direct answer to the only question that matters before committing a year: has this been done? Written to be checked rather than believed — every claim below is one search away from verification.

ORIGINAL IDEA	REVISED CONTRIBUTION	NEAREST PRIOR WORK	DISTANCE FROM IT
Implemented commercially	Open	Feb 2026 preprint	Substantial

1 · Short answer

Both halves of your question, answered

THE ORIGINAL IDEA – SETTLED, DO NOT PRESENT IT AS NEW

"An AI that anticipates upcoming load and pre-adjusts the powertrain" is a shipped commercial product.

Mercedes-Benz Trucks sells Predictive Powertrain Control, which reads GPS and map data to look ahead and manage the powertrain accordingly. Reinforcement learning applied to engine control is also an active published field, including work from Oak Ridge National Laboratory on dilute combustion control and from NREL on injection timing.

If the proposal is presented as "we will build a predictive engine controller," the correct response from a reviewer is that this exists. So we are in the branch where something must be added — and this document is what we added.

THE REVISED CONTRIBUTION – OPEN, AS FAR AS WE CAN ESTABLISH

Nobody has produced a quantitative, empirically validated criterion for how much preview is worth, as a function of system dynamics, across physically different systems. The literature tells you how to *build* a preview controller, and proves that preview benefit converges as the horizon grows. It does not tell an engineer deciding whether to buy map data, add a sensor, or install a communication link whether that investment will pay.

That is the question we now answer, and it is why the second plant exists.

2 · Prior art

What exists, and exactly how close it comes

Ordered by how threatening each is to the claim. Read the last column — it is the one that matters.

PRIOR WORK	WHAT IT ESTABLISHES	THREAT	WHY IT DOES NOT CLOSE THE GAP
H∞ preview control regret bounds preprint, Feb 2026	Proves the benefit of preview <i>converges</i> as horizon $\rightarrow \infty$, with a bound decaying geometrically in the closed-loop spectral radius. This is the closest existing result to ours.	HIGHEST	Linear time-invariant systems only, purely theoretical, and explicitly no decay rate. The bound establishes limiting behaviour, not magnitude. It gives no criterion of the form "preview pays when the horizon exceeds some fraction of a time constant," and it is illustrated on one numerical example rather than characterised across parameter ranges.
MPC horizon selection practice	Engineering rule of thumb that the prediction horizon should cover roughly the open-loop settling time.	HIGH	It is guidance for <i>tuning a controller you have already decided to build</i> , not for deciding whether preview <i>information</i> is worth acquiring. It says nothing about how much performance you gain, and it is folklore rather than a measured law.
Mercedes-Benz Predictive Powertrain Control shipping product	Map-based lookahead controlling cruise speed and gear selection on heavy trucks, for fuel economy.	MEDIUM	Demonstrates that preview control works and is commercially worthwhile — <i>for one system, one objective, one horizon</i> . No published characterisation of how the benefit scales, and it targets speed and gear rather than thermal and durability limits.
RL for engine control ORNL, NREL, and a 2026 survey	Reinforcement learning applied to dilute combustion control and to injection timing, plus a survey of ML for engine calibration.	MEDIUM	Establishes the method is viable in this domain — which supports our feasibility rather than threatening our claim. None of it studies the value of preview information, and none crosses domains.
Predictive energy management for hybrids and EVs	A large literature on route-aware energy management, velocity prediction and eco-driving.	LOW	Each paper optimises one system with one preview scheme. The field has never asked the meta-question of what governs preview value <i>across</i> systems, which is the only question we are asking.

THE HONEST SUMMARY OF THE FIELD

There is a mature body of work on **how to build** preview controllers, and a smaller theoretical result that preview benefit **saturates** for linear systems. There is nothing that measures **how much** preview is worth, in nonlinear systems with hard constraints, in a form an engineer can use before building anything.

3 • The gap

Stated precisely enough to be attacked

Vague gaps are indefensible. This one is narrow on purpose:

No published work provides an empirically validated, dimensionless criterion predicting the magnitude of preview benefit in nonlinear systems with hard constraints and superlinear damage costs, tested across physically distinct plants.

Each clause narrows it. Remove any one and the gap closes — which is what makes it defensible.

Why each clause is load-bearing:

CLAUSE	WHY IT MUST BE THERE
empirically validated	The 2026 preprint gives a theoretical convergence result. We give measured magnitudes. Without this word the preprint has us.
dimensionless criterion	The point is transferability. A result in seconds and kelvin does not move between domains; a result in H/τ does.
nonlinear, hard constraints	All existing preview theory we found is LTI. Knock limits and thermal limits are hard constraints, and damage costs are superlinear — that is where the interesting behaviour lives and where the linear theory is silent.
across physically distinct plants	A single-system result is an anecdote. Combustion and electrochemistry share no physics, so a curve that fits both is evidence of something general.

4 · Contribution

Three claims, in decreasing order of confidence

CLAIM 1 · HIGH CONFIDENCE A measurement method A reproducible procedure for measuring what preview is worth on any plant: matched reactive and predictive controllers with identical actuators, differing only in when they act, with the preview channels ablatable to verify the comparison is clean. Already demonstrated. On the engine: 62 % damage reduction with preview against 32 % without, and an exact collapse onto the reactive policy when preview is zeroed.	CLAIM 2 · THE THESIS H/τ governs the magnitude The value of preview is predicted by the ratio of preview horizon to the time constant of the dominant constrained state — not by domain, actuators or units. Preliminary evidence. Holding policies and physics fixed and sweeping only the time constant moved the preview advantage from 7.6 to 49.9 points — a 7× swing from a dimensionless number alone.	CLAIM 3 · TO BE TESTED It transfers across domains Points from a combustion engine and from an electrochemical battery fall on one curve when plotted against H/τ. Untested. This is what the second plant is for, and it is the claim most likely to fail. Section 6 says what we do if it does.
--	---	---

Claim 1 is already delivered. Claim 2 has preliminary evidence from a controlled sweep. Only Claim 3 is genuinely at risk — and if it fails, Claims 1 and 2 still constitute a complete thesis, with the failure of Claim 3 as a reported finding rather than a collapse.

5 · Objections

The four a reviewer will raise

"The linear preview theory already implies this."

Partly true, and we should concede that immediately. The 2026 regret result proves preview benefit converges with a bound decaying in the closed-loop spectral radius — which is, in effect, the linear-systems shadow of our claim. Our answer has three parts: the bound establishes *that* benefit saturates, not *how much* benefit there is at a given horizon; it is restricted to linear time-invariant systems with no hard constraints, whereas knock limits, thermal limits and superlinear damage costs are exactly where preview behaves interestingly; and it is unvalidated against any physical system. We supply measured magnitudes where the theory supplies existence.

"Mercedes already does this."

They do, on trucks, for speed and gear selection, to save fuel. That is a demonstration that preview control is worth building for one system, one objective and one horizon. It is not a characterisation of how the benefit scales, and it publishes no such analysis. Our question — *would preview pay on a system that is not a truck, and how would you know before building it* — is untouched by a shipping product.

"Why not just run MPC and see?"

That is precisely what the MPC oracle baseline does, and it is the right way to *measure* a single system. But it requires you to already have a validated plant model, a horizon choice and a solver. The criterion's value is that it answers the question *before* that investment, from two numbers an engineer can estimate on a whiteboard. MPC is the measuring instrument; the criterion is the thing being measured.

"An undergraduate project cannot establish a general law."

Correct, and we do not claim one. We claim a criterion supported by two plants, with the aircraft case explicitly labelled as untested extrapolation. The honest framing is *evidence for a criterion*, not *proof of a law*, and the thesis should say so in those words — including in the abstract.

6 · Falsification

What would show we are wrong

A claim that cannot fail is not worth making. Ours fails in three identifiable ways, and each has a prepared response.

IF THIS HAPPENS	THEN
The two plants' curves do not overlap	H/τ is necessary but insufficient. Identify the missing dimensionless group — most likely actuator authority relative to disturbance magnitude — and report the criterion as incomplete with the missing dimension named. <i>Still a contribution.</i>
Preview advantage does not vary with H/τ on the battery	The engine result was a coincidence of that plant's structure. Report the negative result, and re-examine whether the constrained state we chose was really the dominant one.
Someone publishes this first during the year	Cite them, reproduce their result on our two plants, and reframe as independent replication plus extension to constrained nonlinear systems. Replication is publishable and it is honest.

7 · Caveat

What this review does and does not cover

READ THIS BEFORE RELYING ON THE CLAIM

This is a **targeted search, not a systematic review**. It covers preview control theory, predictive powertrain control, and reinforcement learning for engine calibration — the three areas most likely to contain a conflicting result. It does **not** cover process control, robotics, building HVAC, or power systems, any of which could plausibly contain an equivalent criterion under different vocabulary.

Before the proposal is approved, someone should spend a day searching those fields specifically, using terms like *value of information*, *preview gain*, *anticipatory control*, and *lookahead benefit*. If a conflicting result exists, it is far better to find it in week 2 than in the viva — and finding it would not kill the project, it would redirect it toward replication and extension.

Say this to the professor explicitly. A student who states the limits of their own literature search reads as careful; one who claims exhaustive novelty reads as naive, and invites exactly the scrutiny they were trying to avoid.

8 · Sources

Everything above, checkable

- **Regret of H^∞ Preview Controllers** — arXiv preprint 2602.01420 (Feb 2026). Proves H^∞ performance and regret converge as preview steps grow, with a bound decaying in the closed-loop spectral radius. Linear discrete-time systems, theoretical, no explicit decay rate. *The nearest work to ours; cite it and distinguish from it explicitly in Chapter 2.*
- **Mercedes-Benz Trucks Predictive Powertrain Control** — manufacturer documentation of the shipping product. Establishes commercial viability of map-based lookahead for powertrain management.
- **Maldonado, Kaul, Schuman & Young (2024)** — reinforcement learning applied to dilute combustion control for fuel efficiency, Oak Ridge National Laboratory.
- **Henry de Frahan, Wimer, Yellapantula & Grout (2022)** — deep RL for dynamic control of fuel injection timing, NREL.
- **Survey of machine-learning methods for IC engine calibration and real-time control (2026)** — establishes the state of the field; useful for positioning and for showing awareness.

- **Optimal preview control literature** — a substantial body on preview controller synthesis for linear discrete-time systems, including periodic and time-delay variants. Concerned with design, not with quantifying benefit.
 - **MPC horizon selection guidance** — standard practice that the prediction horizon should span the open-loop settling time. Adjacent folklore; cite it as the informal version of what we are trying to measure.
-

What to say in the room, in two sentences. "The idea as originally framed is already a commercial product, so we changed the question. We are not building a predictive controller — we are measuring what preview is worth as a function of system dynamics, which the literature establishes converges but has never quantified in constrained nonlinear systems or tested across domains."

And the sentence that earns trust. "The closest work is a February 2026 preprint on preview control regret bounds; it proves convergence for linear systems without a decay rate, and we can explain precisely how we differ from it."